www.youtube.com/c/javaexpress
javaexpresschannel@gmail.com
+91 7801007910
Java Express

Spring Boot & Microservices Live Sessions Zero to Hero Course

Prerequisites for this Course:

- **Core Java & Java 8 (Java 8 Videos Available for revision)**
- **Spring Basics (Videos Available for revision)**

For Queries:

WhatsApp: +91 7801007910 (Meeting Details)

Email: javaexpresschannel@gmail.com

Course Details:

- **Course Duration :** 7 Weeks (35 Days)
- **Start Date :** 14th Nov @ 9 PM – 10 PM IST
- **Trainer Name :** Venkata Bhaskar
- **Class Schedule:** Mon – Saturday (Saturday is Optional)
- **Backup Videos :** Life Time Access
- **Interview Guide**: Will be provided for your preparation
- **Who Can Take the Course**: Suitable for individuals from Fresher to Experienced (10+ years of experience can also enrol)

Agenda: (Live Sessions) – Zero to Hero Course

- Week 1: Introduction to Spring Boot
- Week 2: Building REST APIS with Spring Boot
- Week 3: Advance Concepts of REST API
- Week 4: Real Time Project – Microservices Development
- Week 5: Microservices Design Patterns
- Week 6: Docker & Kubernetes Deployment for Microservices
- Week 7: Interview Preparation & Q/A
- *Week 8 : Pre-Recorded Videos for Some Design Patterns (CQRS,SAGA,Spring Batch, Rate Limiter)*

www.youtube.com/c/javaexpress
javaexpresschannel@gmail.com
+91 7801007910
Java Express

Additional Free Videos for Devops (Benefit)

- *Week 9 : **Devops** : Pre-Recorded videos (Jenkins Real Time CI/CD Pipeline in AWS for Kubernetes)*

Week 1: Introduction to Spring Boot

Day 1: Introduction to Spring Boot

- Spring Boot Introduction
- Comparison with Spring Framework
- Benefits of using Spring Boot
- Setting up Spring Boot in the IDE
    - Configuring IDE Such as STS & IntelliJ
    - Maven Build Tool for Managing Dependencies

Day 2: Project Creation & Spring Boot Features

- Spring Initializer to create Spring Boot projects
- Project structure and initial configurations
- Building your first Spring Boot application
- Key Files : pom.xml / application.properties / application.yml
- Configuring Properties in application.properties

Day 3: Spring Boot Features & Important Annotations

- Spring Boot Starters
- Embedded Servers
- Exploring auto-configuration (@EnableAutoConfiguration)
- Spring Boot Annotations
    - @SpringBootApplication
    - @Component
    - @Component Scanning
- Working with Beans (@Component)
- Exploring Default Actuator Endpoints

JAVA EXPRESS
Enlight New Thoughts

www.youtube.com/c/javaexpress
javaexpresschannel@gmail.com
+91 7801007910
Java Express

Day 4: Spring Data JPA & Setting up Database

- Introduction to Spring Data JPA
- Benefits of using Spring Data JPA for database operations
- Install and set up a Relational Database (Eg : MYSQL)
- Configuring Data Source in Spring Boot
- Test the configuration by connecting to the database and logs for successful connection

Day 5: Understanding JPA Entities and Repositories

- Create Basic JPA Entity using @Entity, @Id, @GeneratedValue & @Column
- Map Entity fields to database columns
- Create a Repository interface by extending JpaRepository
- Explore commonly used Methods
    - Save
    - findById
    - findAll
    - **delete**

Week 2: Building REST APIS with Spring Boot

Day 6: Introduction to REST APIS

- Overview of REST APIS and how they work
- Understand HTTP Methods (GET,POST,PUT,DELETE)
- Configuring Database Properties in application.properties
- Building Basic CRUD Endpoints
- Basic Package Structure of Real Time Project

Day 7: Implement REST CRUD Operations

- Creating API Endpoints for CRUD Operations
- Controller Setup (@RestController & @RequestMapping)
- Use @GetMapping & @PostMapping to define endpoints
- Understand @RequestBody Annotation to capture the Payload
- Testing API with Postman

www.youtube.com/c/javaexpress
javaexpresschannel@gmail.com
+91 7801007910
Java Express

Day 8: Update & Delete Operations

- Add a PUT Endpoint for updating resources
- Use @PathVariable to capture resource Id a
- Implement Delete Endpoint
- Test the PUT and Delete endpoints using Postman

Day 9: Query Methods & Custom Queries

- OneToMany & ManyToOne Mapping
- Use method naming conventions in Spring Data JPA to derive queries automatically.
- Use `@Query` annotation to define custom JPQL or native SQL queries.
- Create endpoints that use custom queries to filter or retrieve specific data.
- Use Inner Join for Custom Queries & Derived Queries

Day 10: Implementing Pagination and Sorting

- Use Pageable to implement pagination in your API
- Create New Endpoint to accept pagination Parameters (page,size)
- Test pagination using Postman

Week 3: Advance Concepts of REST API

Day 11: Exception Handling in REST APIS

- Use @RestControllerAdvice to handle exceptions globally
- Customize error messages and response codes for different exceptions
- Create Custom exception classes for application specific errors
- Return meaningful error responses when resources aren't found

Day 12: API Documentation with Swagger & Consumer Development

- Generating API Documentation using OpenAPI
- Exploring OpenAPI Annotations
- Overview of Spring Rest Template for making Http Request
- Sending GET,POST,PUT,Delete request

Day 13: Spring Boot Profiles & Audit Fields

- Configuring profiles for different environments.
- Profile-based database configurations and environment-specific settings.
- Adding audit fields to track data changes.

www.youtube.com/c/javaexpress
javaexpresschannel@gmail.com
+91 7801007910
Java Express

Day 14: Spring Boot Unit Testing

- Mockito Unit Testing
- Writing test cases for REST Controllers

Day 15: Mongo DB Integration & DTOS

- Setting up a MongoDB instance (local or Dockerized)
- Introduction to Spring Data MongoDB
- Adding MongoDB dependencies to a Spring Boot project
- MongoDB configurations in `application.properties`
- Implementing CRUD repositories with `MongoRepository`
- Basic CRUD operations with MongoDB in Spring Boot

Week 4: Real Time Project – Microservices Development

Day 16: Introduction to Microservices Architecture & Project

- Understand the differences between monolithic and microservices architectures.
- Learn the advantages and challenges of microservices.
- Real Time Project - **Bank Application Microservices Architecture project**

Day 17-19: Project Setup & Environment

- Create basic CRUD operations for each microservice.
- Implement Swagger documentation for each microservice.
- Set up exception handling for each microservice.
- Use a database of your choice for each microservice.

Week 5-7: Microservices Design Patterns

Day 20: Registry Design Pattern & Client Side Discovery Pattern

- Introduction to Netflix Eureka
- Setting up Eureka Server for service discovery.
- Configuring Eureka Client in microservices.

www.youtube.com/c/javaexpress
javaexpresschannel@gmail.com
+91 7801007910
Java Express

Day 21: Communication between Services (Proxy Design Pattern)

- Feign Client : Setup Inter-service communication between microservices
- Test the services and their endpoints using tools like Postman

Day 22: Cross Cutting Pattern (API Gateway Design Pattern)

- Overview of the API Gateway pattern.
- Set up a basic Spring Boot application with Spring Cloud Gateway dependency.
- Define and configure routes in `application.yml`.
- Set up routing to multiple services, e.g., Accounts and Loans services.
- Test routing by sending requests through the gateway.

Day 23: Resilience Design Pattern *(Circuit Breaker Pattern)*

- Introduction to Circuit Breaker Pattern
- Explore how Circuit Breakers improve reliability.
- Add the Resilience4j dependencies to your Spring Boot project.
- Set up a simple service call to simulate interaction between services.
- Annotate a service method with `@CircuitBreaker` from Resilience4j.
- Configure basic Circuit Breaker properties (failure threshold, wait duration).
- Simulate failures to test the Circuit Breaker's behaviour (Open, Closed, Half-Open states).
- Define FallBack methods to handle errors gracefully when the Circuit Breaker opens.
- Use `@CircuitBreaker` with FallBack logic.

Day 24: Observer Design Pattern

- Introduction to Distributed Tracing
- Setting Up Zipkin for Trace Analysis
- Tracing Requests Across Services

Day 25: Dockerizing Multiple Microservices

- Introduction to Docker and DockerFile
- Building Docker Images for each service
- Pushing Docker Images to Docker Hub

www.youtube.com/c/javaexpress
javaexpresschannel@gmail.com
+91 7801007910
Java Express

Day 26: Docker Compose for Multi-Service Architecture

- What is Docker Compose? How it helps in managing multi-container applications.
- Writing a `docker-compose.yml` file to manage multiple microservices.
- Using `docker-compose up` to start multiple services together.
- Exposing microservices on different ports.

Day 27-28: Kubernetes Introduction & Deployment for All MS

- Overview of Kubernetes
- Why Kubernetes for microservices?
- Setting up a local Kubernetes cluster (e.g., Docker Desktop).
- Understanding Kubernetes Objects
    - Deployment
    - Service
- Introduction to Kubernetes YAML files (manifests) and their importance.
- Basic structure of a manifest file: `apiVersion`, `kind`, `metadata`, `spec`.
- Understand different types of Kubernetes objects that you will define in manifest files (e.g., Pods, Deployments, and Services).
- Hands-on: Create a simple manifest file for deploying a single service.
- Structure of a Deployment manifest file.
- Define the container image and port details for a microservice.
- Hands-on: Create a `deployment.yaml` for one microservice (e.g., Accounts microservice).
- Create a `service.yaml` file to expose the Accounts microservice.

Day 29: Deploying Microservices in Kubernetes Cluster (Using Docker Desktop)

- Create individual `deployment.yaml` and `service.yaml` files for multiple microservices (e.g., Accounts, Loans, Cards).
- Hands-on: Create manifests for multiple microservices and deploy them using `kubectl`.

Day 30: Event-Driven Microservices with Kafka

- Introduction to Kafka for event-driven architecture
- Configuring Kafka with Spring Boot
- Producing and consuming events for Microservices

www.youtube.com/c/javaexpress
javaexpresschannel@gmail.com
+91 7801007910
Java Express

Day 31: Monitoring: **Prometheus and Grafana with Docker Compose**

- Add Prometheus and Grafana for existing docker-compose.yml file
- Expose metrics from microservices for Prometheus to scrape.
- Configure Prometheus to scrape your microservices' metrics by editing the `prometheus.yml` configuration file.
- Install and configure Grafana as a service in the Docker Compose file.
- Connect Grafana to Prometheus as a data source.
- Create basic Grafana dashboards to visualize microservice metrics
- Hands-on: Set up Docker Compose to monitor all microservices and create a combined dashboard in Grafana.

Day 32: Final wrap up

- Q&A session on all Topics
- Interview Preparation

Week 8: Pre-Recorded Videos (CQRS, SAGA, Spring Batch, Rate Limiter)

Distributed Transaction Management - Database Design Patterns (CQRS)

- What is CQRS (Command Query Responsibility Segregation)?
- Components of CQRS: Command side, Query side, and Event handling.
- Introduction to Axios for making HTTP requests.
- Installing and configuring Axios for communicating between services.
- Hands-on: Set up a basic project structure for CQRS implementation.
- How to handle commands in microservices (validation, execution).
- Use Axios to send commands from the client to the backend.
- Hands-on: Create a "Create User" command service and test using Axios
- What is the Query side in CQRS? (Handling Read operations).
- Implementing a query service
- Use Axios to send queries and retrieve data from the Query service.

Database Design Patterns (SAGA) – Ecommerce MS

- Introduction to Orchestration-based SAGA
- Building the Orchestrator Service
- Design and structure of an orchestrator service.
- Using Axios within the orchestrator to call other services in a sequence.
- Implementing Compensating Transactions

www.youtube.com/c/javaexpress
javaexpresschannel@gmail.com
+91 7801007910
Java Express

- Hands-on: Implement a full orchestrator service, handling order fulfilment using Axios for communication.
- Using Axios to trigger compensating transactions in case of failure (e.g., payment refund).
- Hands-on: Implement compensating actions across services using Axios.

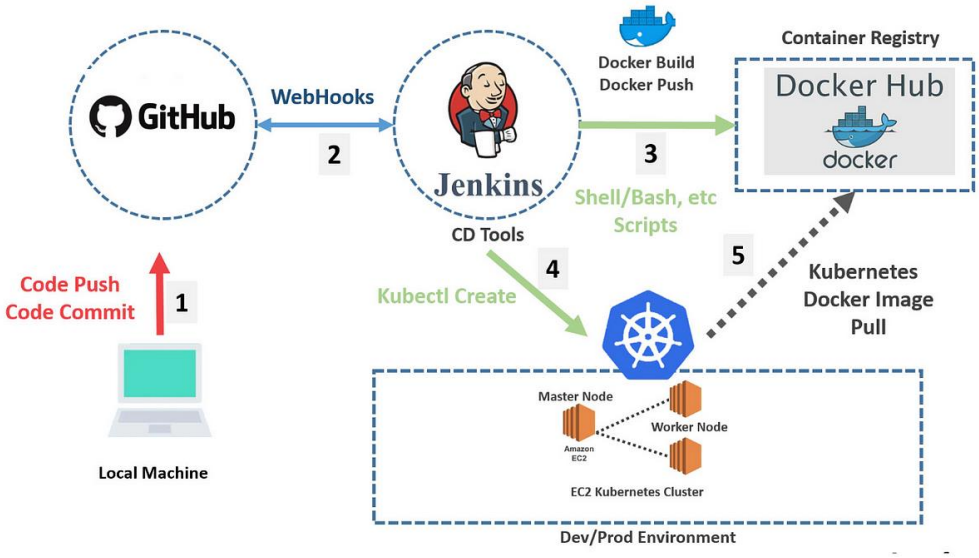
Rate Limiter Design Pattern

- Introduction to Rate Limiting
- Relationship between API Gateways and rate limiting.
- Using Spring Cloud Gateway's **Redis** integration for distributed rate limiting.
- Understanding the **Token Bucket** algorithm.
- How the token bucket works: Tokens refill at a constant rate, and requests consume tokens.
- Hands-on: Implement rate limiting in Spring Cloud Gateway using a chosen algorithm (Token Bucket).

Spring Batch Module

- Introduction to Spring Batch
- Spring Batch Architecture
- Configuring Jobs and Steps
- Item Readers and ItemWriters
- Item Processor and Data Transformation
- Handling Large Datasets and Chunk Processing
- Job Execution, Monitoring, and Logging
- Spring Batch Job Scheduling

Week 9: Pre-Recorded videos (AWS Kubernetes Java Project through Jenkins Deployment CI/CD & Sonar)



www.youtube.com/c/javaexpress
javaexpresschannel@gmail.com
+91 7801007910
Java Express

- **Jenkins Real Time CI/CD Pipeline in AWS for Kubernetes**
  - Jenkins CI/CD
  - Git Integration
  - Maven Integration
  - Docker Integration
  - Sonar Integration
  - Kubernetes Deployment using AWS